SECTION A - TRUE/FALSE QUESTIONS [10 MARKS]

This Section Consists of FIFTEEN (15) Questions.

Write your Answer **TRUE** or **FALSE** for each statement given below in the answer booklet given.

1. Physical DFD depicts the system and describe the business events that take place and the data required and produced by each event. F

2. Developing physical DFD include identifying temporary data stores, specifying actual names of files and printouts and adding controls to ensure the processes are done properly. T

3. Physical data flow diagrams are often more complex than logical data flow diagrams because of the many data stores present in the system. T

4. Decision tables and decision trees are better suited to summarize complex decision logic concisely than structured English in specifying the process. T

5. Decision tree method must consider on actions, decisions, conditions and outcomes of process descriptions. F

6. selection / Branching
An **IF…THEN...ELSE** structured must be presented in the structured English statements when iteration is indicated for an element or a group of elements in the data dictionary. T F

7. Data Couple shows data that one module passes to another meanwhile Control Couple shows a message, also called a flag, which one module sends to another. T

8. Loosely coupled modules are easier to maintain and modify, because the logic in one module does not affect other modules. T

9. Diamond shape in structure chart shows the conditional calling of low-level modules. The program calls modules only when certain conditions exist. T

10. Hidden fields are used on a second form when multiple forms are required to capture all the transaction data. T

11. A report is a business document that contains some predefined data and may include some areas where additional data are to be filled in. T F
report is output not input

12. Ad-hoc reports provide details behind the summary values on a key-indicator or exception report. F

13. Testing managers are responsible for developing test plans, establishing testing standards, integrating testing and development activities in the life cycle, and ensuring that test plans are completed. T

14. The people who create the test cases can be the same people as those who coded and tested the system. T

15. Often organizations that are using cloud computing do not find it necessary to make up-front capital expenditures on IT infrastructure, so it enables smaller companies with smaller and less predictable budgets to make advances in processing more quickly. T

SECTION B - STRUCTURE QUESTIONS [50 MARKS]

This Section Consists of FOUR (4) Questions. Please Answer Question 1 and TWO (2) other Questions.

Write all your answers in the answer booklet provided.

QUESTION 1 (20 MARKS)

Figure 1 shows the physical Data Flow Diagram for Book Rental System. Based on Figure 1 answer the following questions:

a) Partition the physical DFD. Justify your reason for each partition. (Note: Use ***Attachment 1*** to answer this question then attach the page with your name on it into the Answer Booklet).

(6 Marks)

b) Identify at least TWO (2) processes for input design.

- add new customer
- check in book return

(4 Marks)

c) Identify at least TWO (2) processes output design.

- Produce management report
- Produce cash received report

(4 Marks)

d) Based on Figure 1, Physical DFD for Book Rental System, convert the DFD into a structure chart.

(6 Marks)

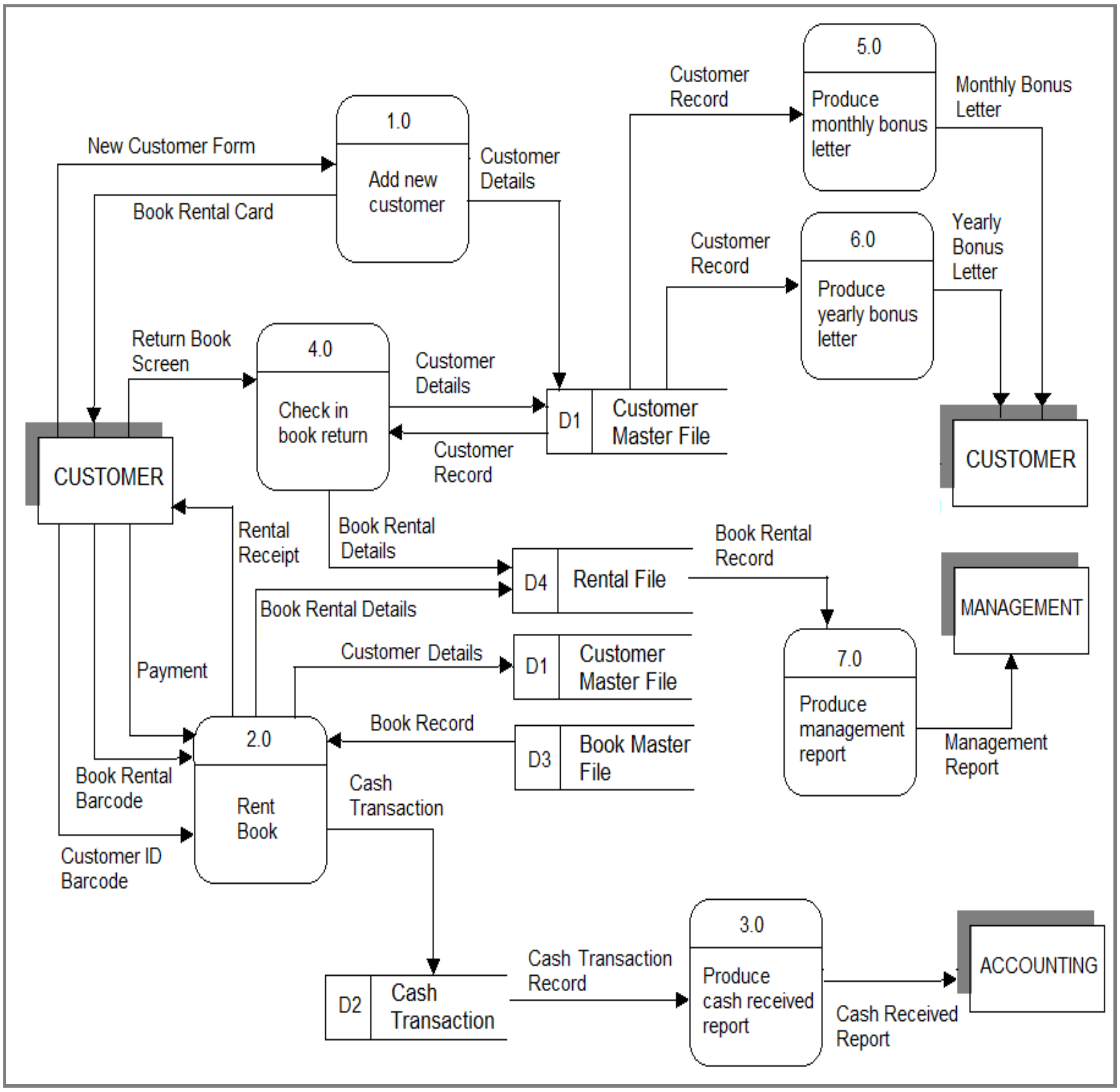


Figure 1: Physical DFD for Book Rental System

QUESTION 2 (15 MARKS)

Your company has been hired as IT consultant to assist 'Exotic Treat' company in developing a newly system that allows their customer to purchase homemade sweets and cakes through online booking application. One of your job as the in-charge Business Analyst is to assist the development team in analyzing the detail process and logic flows for the determination of total payment invoice. The criteria used for the calculation if the invoice are as follows:

- Customers with e-coupon will be granted for RM10 discounts less for each total payment invoice
- Customers who spent more than RM250 purchase items with:
  i) Numbers of order items (more OR less than 4 order items);
  ii) Choices of delivery day once the order is placed (next day, second day, OR seventh day)

The specific conditions to calculate total payment invoice are summarized using the decision table in Table 1.

Table 1: Decision table for calculating shipping charges and total invoice

Total *RMpurchase* **> RM250**	YES						NO					
Number of items (n)	N ≤ 3			N ≥ 4			N ≤ 3			N ≥ 4		
Delivery day	Next	2nd	7th	Next	2nd	7th	Next	2nd	7th	Next	2nd	7th
Delivery *RMcharge*	25	10	n*1.5	n*6.0	n*2.5	Free	35	15	10	n*7.5	n*3.5	n*2.5
e-Coupon	YES						NO					
Total invoice (RM)	(Total*RMpurchase* + Delivery*RMcharge*) – RM10 discounts less						Total *RMpurchase* + Delivery*RMcharge*					

(a) Construct an optimized decision tree to demonstrate the logic conditions and actions sequences of process flows in calculating the total payment invoice

(9 Marks)

(b) In your opinion, which method (decision table or decision tree) is more helpful to analyze the process flows in calculate total payment invoice? Justify your reason.

(6 Marks)

Decision table is more helpful to analyze the process flows in calculate total payment invoice.
This is because all the conditions are relevent to every action, ensuring the completeness of every condition affects the 'Delivery RM charge' differently

QUESTION 3 (15 MARKS)

Design one sample data entry screen for Homestay Reservation System using the entry guideline provided in Table 2 below. Support your design with arguments for each of the design choices you made.

(15 Marks)

Table 2: Guidelines for Structuring Data Entry Fields

Entry	Never require data that are already online or that can be computed; for example, do not enter customer data on an order form if those data can be retrieved from the database, and do not enter extended prices that can be computed from quantity sold and unit prices.
Defaults	Always provide default values when appropriate; for example, assume today's date for a new sales invoice, or use the standard product price unless overridden.
Units	Make clear the type of data units requested for entry; for example, indicate quantity in tons, dozens, pounds, etc.
Replacement	Use character replacement when appropriate; for example, allow the user to look up the value in a table or automatically fill in the value once the user enters enough significant characters.
Captioning	Always place a caption adjacent to fields
Format	Provide formatting examples when appropriate; for example, automatically show standard embedded symbols, decimal points, credit symbol, or dollar sign.
Justify	Automatically justify data entries; numbers should be right justified and aligned on decimal points, and text should be left justified.
Help	Provide context-sensitive help when appropriate; for example, provide a hot key, such as the Fl key, that opens the help system on an entry that is most closely related to where the cursor is on the display.

<u>QUESTION 4 (15 MARKS)</u>

a) There are many conversion strategies available to analysts, and also a contingency approach that takes into account several user and organizational variables in deciding which conversion strategy to use. There is no single best way to proceed with conversion. The importance of adequate planning and scheduling of conversion with the strategic involvement of users (which often takes many weeks), file backup, and adequate security cannot be overemphasized. List the **FOUR (4)** conversion strategies for converting old information systems to new ones.

(2 Marks)

(2) Describe briefly TWO of conversion strategy in (a) above using a diagram to demonstrate your understanding.

(4 Marks)

(3) Define the terms physical, logical, and behavioral security, and give an example of each one that illustrates the differences among them.

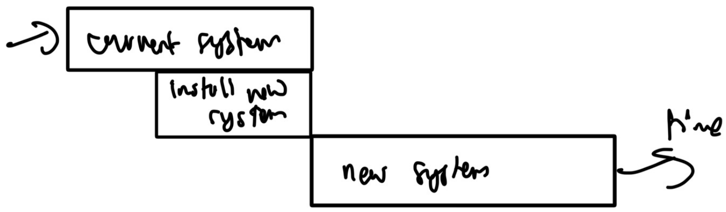
(4 Marks)

(4 List TWO (2) of the several measures an analyst can take to improve the security, privacy, and confidentiality of data, systems, networks, individuals, and organizations that use ecommerce Web applications.

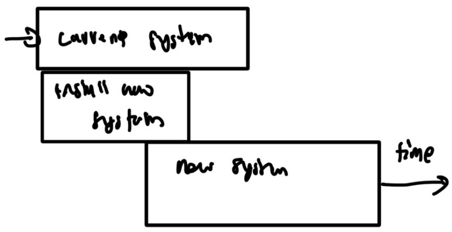
(5 Marks)

a) - Direct
- parallel
- single location
- phases

i) Direct installation
- changing over the old system to a new one by turning off the old system when new one is turned on



ii) Parallel installation



SECTION C – CASE STUDY [40 MARKS]

This Section Consist of ONE (1) Question.

PLEASE ANSWER THESE QUESTIONS in the Answer Booklet

QUESTION 1 (40 Marks)

Mobile Book Café (MBC) Case Study

Introduction

Mobile Book Café (MBC) is a book shop based on the ground floor of a shop house in Kajang Central. Currently, MBC has about 15,000 titles in stock and on the shelves, where they are organized in subject groupings. The problem in the shop is to organize the book order in a systematic way. The current process of ordering book is described below. After choosing book(s), customer will take the book(s) to the counter.

Current System

During weekend there are many customers in the shop. When books are sold over the counter, the sales person records the ISBN of each book, together with its price. (ISBN, International Standard Book Number, is a unique number for each book published). At the end of each day, the list of books sold is produced and is checked against the current stock.

The process involve in order book from book's vendor are:

1. Check the books that are sold from the file **SOLD OUT.**
2. Check the balance of the book with the current stock and make a list of book need to be ordered and put in the file **NEED ORDER.**
3. Order through telephone, mail, order form or catalogue from book's vendor and keep in the file **ORDER.**

New System

MBC wants to build a new system that can handle customers' order and vendors order. The system must run on the internet. The information about the book that MBC can sell over the internet is similar to the book's database at any retail store (e.g. title, ISBN, quantity in stock). The customer can order or search information of books through internet. To search book's information, customers need to give the book's title and its ISBN number. To place an order, a customer has to register with the system. Payment of the order is through credit card, system will need to verify the customer's credit card number. At the end of the day, the system will produce a report of books sold for the day and balance from the stock. Using this list, the manager will then reorder the books from the vendor through the electronic mail.

Based on the above Mobile Book Café Case study,

a) List all the external entities

(5 Marks)

b) List all business processes

(5 Marks)

c) Draw a context diagram for the internet book sales system.

(10 Marks)

d) Explode the context-level diagram in Question (c) showing all the major processes. Call this Diagram ‘0’. It should be a logical data flow diagram.

(10 Marks)

e) Design ONE (1) user interface for “Book Searching” process in the new system

(5 Marks)

f) Suggest the implementation plan for Mobile Book Café to change from current systems to a new system

(5 Marks)